

DEVA ANAND

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Education

MS, Computer Science, University of Massachusetts Amherst 2025–2027
Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering

BE, Computer Science and Engineering, Vels Institute of Science, Technology and Advanced Studies 2019–2023

Professional Experience

TherAlign Health, Founding Software Engineer Intern | *Remote* Jul 2026 – Present

- Built provider-facing **SMART on FHIR** decision-support workflows using **Next.js** and Firebase/**GCP** cloud services, surfacing guideline-backed medication alternatives during prescribing to reduce prior-authorization friction.
- Extended a **Cloud Functions** pipeline combining **RxNorm** normalization, formulary rules, and PubMed evidence; restricted **Gemini** to evidence synthesis while eligibility and coverage remained deterministic.

Culture & Morality Lab, UMass Amherst, Lab Assistant | *Amherst, MA* Jul 2026 – Present

- Sole engineer building the lab's CCR text-analysis platform (Python, **FastAPI**, sentence-transformers, React) under full technical ownership, productionizing a published NLP method into a self-serve web tool for researchers.
- Built a YAML model registry, a versioned construct library (**99 constructs from 38 questionnaires**), and a structured data-quality warning system; grew the backend suite to **40 tests** (hermetic CI).

Wysa, Backend Engineer | *Bengaluru, India* Jul 2023 – Jul 2025

- Built scalable cloud backend services and APIs for a multi-tenant NHS eTriage system (Node.js, MongoDB) serving **8+ UK clinical clients** and **10,000+ monthly submissions** via Mayden iaptus REST integrations; contributed to **\$340K+** in revenue.
- Designed and coded a full-stack React web portal used daily by the AI team to structure and label ML training data, cutting manual annotation by **100+ hours/month** and lifting labeling throughput by **60%**.
- Owned backend security and reliability for the eTriage stack, implementing **AWS KMS** encryption for clinical PII, hardening authentication and rate limiting, and remediating **20+ VAPT findings** with regression-tested fixes.
- Served as primary technical point of contact for the eTriage backend; ran structured knowledge transfer that onboarded a second engineer to on-call readiness in **~6 weeks**, and piloted a COO-sponsored GitHub Copilot productivity experiment.

Projects

AgenticSearch – Provenance-First Entity Discovery (GitHub | Demo) Python, FastAPI, OpenAI / Groq, SQLite

- Built a cloud-hosted Python/FastAPI backend service that uses ML/AI to turn open-ended queries into structured, verifiable entity tables via typed retrieval planning, reranking, and deterministic extraction with LLM fallback; hardened with **150+ automated tests**.

cf-ai-research-scout – Edge-Native AI Assistant (GitHub) TypeScript, Cloudflare Workers AI, Durable Objects, D1

- Built a full-stack edge web app with a chat UI over serverless cloud services (Cloudflare Workers AI, Durable Objects, D1), streaming AI responses in real time with persistent per-session state and a typed client-server API.

Spark ETL Performance Optimization (GitHub) PySpark, Spark UI, SQL, NYC TLC Dataset

- Optimized a large-scale analytics pipeline over **~1.4B rows**, cutting end-to-end runtime **~25%** and the tail-latency ratio **2.14x to 1.74x** via broadcast joins, AQE, skew handling, and partitioning, profiled via Spark UI.

Publications

- "Alzheimer's Disease Classification using Transfer Learning," IEEE CONIT 2023, first author, deep transfer learning on neuroimaging data. (Paper)

Skills

Languages & Databases: JavaScript, TypeScript, Python, C++, Java, SQL, Bash, PostgreSQL, MongoDB, MySQL, SQLite

Frameworks / Tools: Node.js, Express, React, Next.js, FastAPI, Docker, Cloudflare Workers, Git, REST APIs

DevOps / Testing: Linux, shell scripting, CI/CD, GitHub Actions, unit & integration testing, monitoring, automation

ML & AI: PyTorch, Hugging Face Transformers, sentence-transformers, LLM integration, RAG, analytics pipelines, evaluation

Cloud / Platforms: AWS, GCP, Azure, Firebase, MongoDB Atlas

Open Source & Community

- Built **Underdogs Fitness** (Website), a production MERN platform, solo as a freelance engagement; live **3+ years**. - Selected for **GirlScript Summer of Code 2023**; contributed to Linkfree, Freehit, and ProjectsHut. - Tutored **30+ underprivileged students** in programming at Sayur, a non-profit in Tamil Nadu.